Arab Academy for Science & Technology & Maritime Transport
College of Engineering & Technology
Department of Basic & Applied Science

Mathematics 2

Sheet

Arab Academy for Science & Technology
&Maritime Transport
College of Engineering & Technology
Department of Basic & Applied Science

Syllabus for Math (2) - BA 124

Week	Sheet	Topics	Lec	Sec.	H.W.
1	1 1	Definition of indefinite integrals Table of famous integrals	1-10	1-8	1-5
2	1 1	Simple rules of integration Fundamental theorem of calculus	11-23	11-22	6-28
3	1 2	Fundamental theorem of calculus Integration by parts	24-29 1-5	23-33 1-6	29-34 1-4
4	2 2	Integration by parts Integration of rational functions	6-10 12-13	7-12 12-13	5-12 13-14
5	2	Integration of rational functions	14-21	14-22	15-26
6	3	Integration of trigonometric powers	1-12	1-12	1-9
7	3	Trigonometric substitution	13-21	13-18	20-25
8	3 3	Integration of quad. Forms Reduction formulas	18-29 30-34	18-30 31-37	9-19,26-27 28-36
9	4	Definite integration	1-4	1-4	1-7
10	4	Area & Volumes	5-9	5-9	8-13
11	4 4 4	Length of a curve Average of a function Numerical integration	16-18 19-20 21-23	15-17 18-20 21-23	21-24 25-28 29-32
12	5	Matrix algebra	Section A	Section A	Section A
13	5	Solution of systems of linear equations Through : Gauss, Cramer, inverse matrix methods	Section B	Section B	Section B
14	5	Eigenvalues, Eigenvectors and Cayley-Hamilton theorem	Section C	Section C	Section C
15		General Revision			

Table 1

N.B: All the following rules should be known by heart.

Some famous integrals:

1. $\int x^n \quad dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$
2. $\int \frac{dx}{x} = \ln|x| + c$
3. $\int \sin x \quad dx = -\cos x + c$
4. $\int \cos x \quad dx = \sin x + c$
5. $\int \sec x \quad \tan x \quad dx = \sec x + c$
6. $\int \cos ecx \quad \cot x \quad dx = -\cos ecx + c$
7. $\int \sec x \quad dx = \ln|\sec x + \tan x| + c$
8. $\int \cos ecx \quad dx = -\ln|\cos ecx + \cot x| + c$
9. $\int \tan x \quad dx = \ln|\sec x| + c = -\ln|\cos x| + c$
10. $\int \cot x \quad dx = \ln|\sin x| + c = -\ln|\cos ecx| + c$
11. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$
12. $\int \frac{dx}{a^2 - x^2} = \frac{1}{a}\tanh^{-1}\frac{x}{a} + c$
13. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a} + c$
14. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\frac{x}{a} + c$
15. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\frac{x}{a} + c$
16. $\int \tan^2 x \quad dx = \int (\sec^2 x - 1)\, dx = \tan x - x + c$
17. $\int \cot^2 x \quad dx = \int (\cos ec^2 x - 1)\, dx = -\cot x - x + c$
18. $\int \cos^2 x \quad dx = \frac{1}{2}\int 1 + \cos 2x\, dx = \frac{x}{2} + \frac{\sin 2x}{4} + c$
19. $\int \sin^2 x \quad dx = \frac{1}{2}\int 1 - \cos 2x\, dx = \frac{x}{2} - \frac{\sin 2x}{4} + c$
20. $\int \sec^2 x \quad dx = \tan x + c$
21. $\int \cos ec^2 x \quad dx = -\cot + c$
22. $\int e^x dx = e^x + c$
23. $\int a^x dx = \frac{a^x}{\ln a} + c$

Trigonometric formulas:

$\sin^2 x + \cos^2 x = 1$

$\tan^2 x + 1 = \sec^2 x$

$\sin 2x = 2\sin x \cos x$

$$\cos 2x = \cos^2 x - \sin^2 x$$
$$= 2\cos^2 x - 1$$
$$= 1 - 2\sin^2 x$$

$$\mathbf{\cos A \cos B = \frac{1}{2}\left[\cos(A-B) + \cos(A+B)\right]}$$

$$\mathbf{\sin A \sin B = \frac{1}{2}\left[\cos(A-B) - \cos(A+B)\right]}$$

$$\mathbf{\sin A \cos B = \frac{1}{2}\left[\sin(A-B) + \sin(A+B)\right]}$$

Some useful rules:

$a^2 - b^2 = (a+b)(a-b)$

$x^2 \pm 2ax = (x \pm a)^2 - a^2$

If $\quad ax^2 + bx + c = 0$

then $\quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Definition of indefinite integration:

$$\boxed{\int f(x)\ dx = F(x)+c, \qquad where \quad f(x) = F'(x) = \frac{d}{dx}F(x)}$$

Properties of integration:

i) $\int c\ f(x)\,dx = c\int f(x)\ dx$

ii) $\int f(x) \pm g(x)\,dx = \int f(x)\ dx \pm \int g(x)\,dx$

Where c is a constant

Some basic rules:

Rule 1: $\int u^n\ u'\,dx = \int u^n du = \frac{u^{n+1}}{n+1}+c \qquad , where \quad n \neq -1$

Rule 2: $\int \frac{u'}{u}\,dx = \int \frac{du}{u} = \ln u + c$

Rule 3: $\int \frac{u'}{\sqrt{u}}\,dx = \int \frac{du}{\sqrt{u}} = 2\sqrt{u}+c$

Sheet#1
Lecture Examples

Section A:

1. $\int x + x^2 + 3 \quad dx$

2. $\int \frac{dx}{\sqrt[4]{x}}$

3. $\int (x^2 + 2)^2 dx$

4. $\int (\frac{1}{x^2} + \frac{4}{x\sqrt{x}} - 2)\, dx$

Section B: Integration by substitution

5. $\int \frac{x}{\tan x^2} dx$

6. $\int \sec^2 3x \quad dx$

7. $\int (e^{2x})^2 dx$

8. $\int e^{4x} + 5^x dx$

9. $\int \frac{1}{x} \cos(\ln x)\, dx$

10. $\int \frac{\sec\sqrt{x}}{\sqrt{x}} dx$

---

11. $\int (7x^2 - 4)^6 \; x\, dx$

12. $\int \frac{x^2}{20x^3 + 4} dx$

13. $\int \frac{1 - 2x^3}{\sqrt{2x - x^4 + 1}} dx$

14. $\int \tan^3 7x \quad \sec^2 7x \quad dx$

15. $\int \cos^6 2x \quad \sin 2x \quad dx$

16. $\int \sec^8 x \quad \tan x\, dx$

17. $\int \tan x \quad \cos^5 x \quad dx$

18. $\int \frac{dx}{x\,(1 + \ln x)^3}$

19. $\int \frac{(1 + \cot x)^3}{1 - \cos^2 x} dx$

20. $\int (3 + 4\cos x)^2 dx$

21. $\int \frac{1 + \sin x}{\cos^2 x} dx$

22. $\int \frac{1 - \cos 2x}{\sin^2 x} dx$

23. $\int \frac{dx}{x + x\ln x}$

Section C:

24. $\int \frac{dx}{x^2 + 9}$

25. $\int \frac{dx}{\sqrt{25 - 16x^2}}$

26. $\int \frac{dx}{x\sqrt{4 - 9\ln^2 x}}$

27. $\int \frac{e^x + e^{2x}}{e^{2x} + 1} dx$

28. $\int \frac{\cos x}{\sqrt{4 - \sin^2 x}}$

29. $\int \frac{dx}{x(1 + \ln^2 x)}$

Sheet#1
Classroom Exercises

Section A:

1. $\int x + \sqrt{x} - 2 \, dx$

2. $\int (x^2 + \frac{1}{\sqrt{x}})^2 \, dx$

Section B: Integration by Substitution

3. $\int \cos 5x \quad dx$

4. $\int \frac{x \quad dx}{\sin^2 5x^2}$

5. $\int \frac{1}{x} \sec^2 \ln x \; dx$

6. $\int \cot e^x \quad e^x \quad dx$

7. $\int e^{\sin x} \cos x \, dx$

8. $\int (e^{2x} + e^{-2x})^2 \, dx$

---

9. $\int (3x-2)^{10} \, dx$

10. $\int \frac{dx}{5-2x}$

11. $\int \frac{x \, dx}{\sqrt[4]{20x^2+1}}$

12. $\int \cos^4(9x+6) \sin(9x+6) \, dx$

13. $\int \sec^2(7x+3) \tan^2(7x+3) \, dx$

14. $\int \frac{\cos x - \sin x}{\cos x + \sin x} dx$

15. $\int \frac{1}{x} \ln x^2 \, dx$

16. $\int \tan x \; \ln|\cos x| \; dx$

17. $\int \frac{\tan^3 x}{\cos^2 x} dx$

18. $\int \frac{\sqrt{1+\ln x}}{x} dx$

19. $\int \frac{\sqrt{8+\tan x}}{1-\sin^2 x} dx$

20. $\int \frac{\tan^{-1} x}{1+x^2} dx$

21. $\int \frac{\sin 2x}{\sqrt{1+\sin^2 x}} dx$

22. $\int \frac{\sin 2x}{\sin^2 x} dx$

23. $\int \frac{\cos 2x}{\cos x + \sin x}\, dx$

24. $\int (\sin x + \cos x)^2\, dx$

25. $\int (x+1)^3\, (x-1)^3\, x\, dx$

26. $\int (x^2 - \frac{1}{x})^6\, x^8\, dx$

Section C:

27. $\int \frac{dx}{4 + x^2}$

28. $\int \frac{dx}{1 + 9x^2}$

29. $\int \frac{dx}{\sqrt{16 - 9x^2}}$

30. $\int \frac{e^x\, dx}{\sqrt{1 - e^x}}$

31. $\int \frac{\cos x \quad dx}{a^2 + \sin^2 x}$

32. $\int \frac{dx}{x\sqrt{4 - 9\ln^2 x}}$

33. $\int \frac{x - \sqrt{\sin^{-1} x}}{\sqrt{1 - x^2}}\, dx$

Homework#1

1. $\int \frac{1}{\sqrt{x}} + \frac{4}{x\sqrt{x}} - 2\, dx$

2. $\int e^{5x} \quad dx$

3. $\int \frac{dx}{\cos^2 7x}$

4. $\int \frac{Sec^2(\ln x)}{x} dx$

5. $\int \cot(5x-7)\, dx$

6. $\int \tan x \sec^2 x\, dx$

7. $\int \frac{\ln x}{x} dx$

8. $\int \sqrt{x^2+1}\, x \quad dx$

9. $\int (\tan 4z - \cot \frac{z}{4}) dz$

10. $\int \frac{x^2 \quad dx}{\sqrt{x^3+1}}$

11. $\int \frac{\cos x}{\sin^2 x} \quad dx$

12. $\int (\sec x + \tan x)^2\, dx$

13. $\int \frac{dx}{x\sqrt{1-\ln^2 x}}$

14. $\int \frac{dx}{\cos^2 x \sqrt{\tan x - 1}}$

15. $\int \frac{dx}{\sin^{-1} x \quad \sqrt{1-x^2}}$

16. $\int \frac{dx}{x \ln x \quad \ln \ln x}$

17. $\int \frac{\sin^2 x}{1+\cos 2x} dx$

18. $\int \frac{\sin 2x}{\sqrt{1+\sin^2 x}} dx$

19. $\int a^{x^2} x \quad dx$

20. $\int e^{(x^2+4x+3)}(x+2) dx$

21. $\int \frac{x}{(1+3x^2)^{5/2}} dx$

22. $\int \frac{dx}{x \ln x^2}$

23. $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

24. $\int \sec(7x-2)\, dx$

25. $\int \frac{\sec^2 x}{(5\tan x - 3)^6} dx$

26. $\int \frac{1+\tan^2 x}{\tan x} dx$

27. $\int \cos ec^2(\pi x) \quad dx$

28. $\int e^x (e^x + 3)^5 \; dx$

29. $\int \frac{dx}{\tan 3x \quad \sin 3x}$

30. $\int \frac{dx}{\sqrt{9 - x^2}}$

31. $\int \frac{e^x \, dx}{1 + e^{2x}}$

32. $\int \frac{3x+2}{25x^2+49} dx$

33. $\int \frac{dx}{x + x \ln^2 x}$

34. $\int \cos^4 x + 2\cos^2 x \sin^2 x + \sin^4 x \quad dx$

Sheet#2

Lecture Examples

Section A: Integration by parts

1. $\int x \quad \cos x \, dx$
2. $\int \ln x \quad dx$
3. $\int x \quad \sec^2 x \quad dx$
4. $\int e^{(x+2)}(4x+3)\, dx$
5. $\int \ln(x^2+1) \quad dx$
6. $\int x^2 \sin x \quad dx$
7. $\int x^3 \quad (\ln x)^2 \quad dx$
8. $\int \ln^2 x \quad dx$
9. $\int e^{2x} \quad \cos x \quad dx$
10. $\int \sin(\ln x) \quad dx$

Section B: Integration of rational functions

Long division:

12. $\int \dfrac{x^5+8x^3+16x}{x^2+4} dx$
13. $\int \dfrac{2x^3+x^2+9x+4}{x^2+4} \quad dx$

Partial Fractions:

14. $\int \dfrac{dx}{(x-1)(x+1)}$
15. $\int \dfrac{4x^2+6x-30}{(x+1)(x-3)(x+5)} dx$
16. $\dfrac{x+3}{(x+2)(x+1)^2} dx$
17. $\int \dfrac{x}{(x-1)(x+1)^2} dx$
18. $\int \dfrac{dx}{x^3-2x^2+x}$
19. $\int \dfrac{x^2+3x-3}{(x+1)(x^2+4)} dx$
20. $\int \dfrac{5x^3+28x^2+47x-30}{(x^2+9)(x+3)(x+2)} dx$
21. $\int \dfrac{9+7x+x^2}{(1+x)^3(x+2)} dx$

Sheet#2

Classroom Exercises

Section A: Integration by parts

1. $\int x \sin 3x \, dx$
2. $\int x e^{-4x} \, dx$
3. $\int x \ln x \, dx$
4. $\int x \sec x \tan x \, dx$
5. $\int x^4 \ln x^4 \, dx$
6. $\int \frac{\ln(\ln x)}{x} \, dx$
7. $\int x^2 e^{5x} \, dx$
8. $\int x^6 (\ln x)^2 \, dx$
9. $\int \frac{\ln \ln x}{x} \, dx$
10. $\int e^x \sin x \, dx$
11. $\int \sec^3 x \, dx$
12. $\int \cos(\ln x) \, dx$

Section B: Integration of rational functions

Long division:

12. $\int \frac{x^5 - 1}{x^2 - 1} dx$
13. $\int \frac{2x^3 - x^2 + 3x}{x^2 + 1} \, dx$

Partial Fractions:

14. $\int \frac{x^2 + 3x + 4}{(x-1)(x-2)(x-3)} dx$
15. $\int \frac{x + 5}{(x+1)(x+2)^2} dx$
16. $\int \frac{x^2 + 4x + 1}{x^2 + x} dx$
17. $\int \frac{4}{x^2 (x+2)^2} dx$
18. $\int \frac{x^2 - 1}{(x+1)^3} \, dx$
19. $\int \frac{2x - 1}{(x-1)\ (x-2)} dx$
20. $\int \frac{x^2 + 5x + 1}{(x^2 + 1)(x + 2)} dx$
21. $\int \frac{2x^3 - x^2 + 6x - 3}{x^4 - 1} dx$
22. $\int \frac{2x^3 + 3x^2 - x - 2}{(x-1)(x+2)(x^2 + 4)} dx$

Homework#2

1. $\int x \sec^2 x \quad dx$

2. $\int \tan^{-1} x \quad dx$

3. $\int \ln(x-1) \quad dx$

4. $\int x \cos^2 x \quad dx$

5. $\int x^2 e^{5x} \quad dx$

6. $\int x^2 \cos x \quad dx$

7. $\int (x+4)\, e^{2x-3} \quad dx$

8. $\int e^x \sin 3x \quad dx$

9. $\int x^4 \ln x^4 \quad dx$

10. $\int x \tan^2 x \quad dx$

11. $\int x\, 3^x \quad dx$

12. $\int e^{2x} \ln e^{5x} \quad dx$

---

13. $\int \frac{x^7+3x^5+3x^3+x}{x^2+1} dx$

14. $\int \frac{3x^3+2x^2+13x+9}{x^2+4} dx$

15. $\int \frac{dx}{(x-1)(x-2)(x-3)}$

16. $\int \frac{x \quad dx}{(x+1)(x+2)(x+5)}$

17. $\int \frac{3x-5}{(x+3)(x-7)} \quad dx$

18. $\int \frac{4x}{(x-1)(x^2+1)} dx$

19. $\int \frac{x^5+x^4-8}{x^3-4x} dx$

20. $\int \frac{4x^2-8x+4}{(x-1)^3 \quad (x+3)} dx$

21. $\int \frac{x^2+22x-2}{(x+5) \quad (x^2+4)} dx$

22. $\int \frac{4x}{x^4-1} dx$

23.* $\int \frac{3x^4-x^3-2x^2+6}{x^2(x-1)} \quad dx$

24. $\int \frac{x^2+5x}{x^2-4} dx$

25. $\int \frac{dx}{(x^2-x)(x^2-2x+1)}$

26. $\int \frac{2x^2+2x+2}{(x+1)^2(x^2+1)} dx$

Sheet#3
Lecture Examples

Section A: "Trigonometric powers"

a) Integrals of the form $\int \sin^n x \quad \cos^m x \quad dx$

1. $\int \sin^3 x \quad \cos^{-5} x \quad dx$
2. $\int \sin^{\frac{2}{3}} x \quad \cos^3 x \quad dx$
3. $\int \cos^4 x \quad \sin^2 x \quad dx$
4. $\int \cos 3x \quad \cos 7x \quad dx$
5. $\int \cos^3 x \quad dx$
6. $\int \sin^4 2x \quad \cos^4 2x \, dx$
7. $\int \cos^2 x \quad \sin 4x \quad dx$
8. $\int \cos^2 x \quad \sin^2 x \quad dx$
9. $\int \sin 3x \quad \cos 2x \quad dx$

b) Integrals of the form $\int \tan^n x \quad \sec^m x \quad dx$

10. $\int \tan^3 x \quad Sec^3 x \quad dx$
11. $\int \tan^6 x \quad \sec^4 x \quad dx$
12. $\int \tan^5 x \quad \sec x \quad dx$
13. $\int \sec^4 2x \quad dx$
14. $\int \sec^4 x \quad \tan^2 x \quad dx$
15. $\int \frac{\tan^3 dx}{\cos^3 x}$

Section B: "Trigonometric substitution"

Prove the following:

16. $\int \sqrt{a^2 - x^2}\, dx \quad = \frac{a^2}{2} \sin^{-1} \frac{x}{a} + \frac{x}{2} \sqrt{a^2 - x^2} + c$

17. $\int \sqrt{x^2 - a^2}\, dx \quad = \frac{x\sqrt{x^2 - a^2}}{2} - \frac{a^2}{2} \ln \left| \frac{x + \sqrt{x^2 - a^2}}{a} \right| + c$

18. $\int \sqrt{x^2 + a^2}\, dx \quad = \frac{x\sqrt{x^2 + a^2}}{2} + \frac{a^2}{2} \ln \left| \frac{x + \sqrt{x^2 + a^2}}{a} \right| + c$

Solve the following integrals:

19. $\int \frac{\sqrt{x^2-9}}{x} \quad dx$

20. $\int \frac{x^2}{\sqrt{3-x^2}} \quad dx$

21. $\int \frac{dx}{\left(x^2+4\right)^{\frac{3}{2}}}$

22. $\int \frac{x^3}{\sqrt{9+x^2}} \quad dx$

23. $\int x^3\sqrt{4+x^2} \quad dx$

24. $\int x^2\sqrt{1-x^2} \quad dx$

25. $\int \sqrt{8+2x-x^2} \quad dx$

26. $\int \frac{\sqrt{4-\sin^2 x}}{\sec x}\, dx$

Section C:

Integrals of the form $\int \frac{\mathbf{Ax+B}}{\mathbf{ax^2+bx+c}}\mathbf{dx}$ **and** $\int \frac{\mathbf{Ax+B}}{\sqrt{\mathbf{ax^2+bx+c}}}\mathbf{dx}$

27. $\int \frac{dx}{4x^2+16x+12}$

28. $\int \frac{dx}{\sqrt{3x^2-16}}$

29. $\int \frac{dx}{\sqrt{1+x+x^2}}$

30. $\int \frac{x+2}{\sqrt{x^2+2x+3}}dx$

31. $\int \frac{x}{x^2+6x+15}dx$

32. $\int \frac{x^2-1}{\sqrt{x}+2x+1}dx$

33. $\int \frac{dx}{\sqrt{x^2+4x+3}}$

34. $\int \frac{dx}{\sqrt{x(3x+5)}}$

35. $\int \frac{dx}{8+2x-x^2}$

36. $\int \frac{dx}{x\sqrt{ln^2 x-4\,ln\,x+3}}$

37. $\int \frac{\cos x\, dx}{\sin^2 x-4\sin x+12}$

38. $\int \frac{10^x}{10^{2x}-8(10^x)+12}dx$

Section D: Reduction Formulas.

Prove the following reduction formulas:

39. $\int \sin^n x \, dx = I_n = \frac{-1}{n} \sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2}$

And hence find: $\int \frac{\sin^4 x}{\operatorname{cosec}^2 x} \, dx$

40. $\int \cos^n x \, dx = I_n = \frac{1}{n} \sin x \cos^{n-1} x + \frac{n-1}{n} I_{n-2}$

And hence find: $\int \cos^3 x \, dx$

41. $\int \tan^n x \, dx = I_n = \frac{1}{n-1} \tan^{n-1} x - I_{n-2}$

And hence find: $\int \tan^7 x \cot^2 x \, dx$

42. $\int \sec^n x \, dx = I_n = \frac{1}{n-1} \tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2}$

And hence find: $\int \sec^5 x \, dx$

Sheet#3

Classroom Exercises

Section A: "Trigonometric powers"

a) Integrals of the form $\int \sin^n x \quad \cos^m x \quad dx$

1. $\int \sin^2 x \quad \cos^5 x \quad dx$
2. $\int \sin^4 x \quad \cos^2 x \quad dx$
3. $\int \cos^7 x \quad \sin^3 x \quad dx$
4. $\int \sin^4 x \quad \cos^4 x \quad dx$
5. $\int \sin 3x \quad \sin 5x \quad dx$
6. $\int \sin^3 x \quad \cos^{\frac{3}{5}} x \quad dx$
7. $\int \frac{\cos^3 x}{\sqrt{\sin x}} dx$
8. $\int \sin 2x \quad \cos 3x \quad dx$
9. $\int \sin^2 x \quad \cos 4x \quad dx$

b) Integrals of the form $\int \tan^n x \quad \sec^m x \quad dx$

10. $\int \tan^3 x \quad \sec^4 x \quad dx$
11. $\int \tan^3 x \quad \sec^5 x \quad dx$
12. $\int \tan^4 x \quad \sec^4 x \quad dx$
13. $\int \sec^4 x \quad dx$
14. $\int \frac{\sec^3 dx}{\cot^3 x}$
15. $\int \sec x \quad \tan^2 x \quad dx$

Section B: "Trigonometric substitution"

Solve the following integrals:

16. $\int \frac{x^2}{\sqrt{1-x^2}} \quad dx$
17. $\int \frac{x^4}{(x^2+1)^2} \quad dx$
18. $\int \frac{dx}{x^2\sqrt{1-x^2}}$
19. $\int \frac{dx}{x\sqrt{9+x^2}}$
20. $\int \frac{x^5}{\sqrt{1+x^2}} \quad dx$
21. $\int \frac{x^3}{(x^2-4)^2} \quad dx$

22. $\int \sqrt{2-4x+x^2} \quad dx$

23. $\int \sqrt{x^2+4} \quad dx$

24. $\int \sqrt{7-x^2+6x} \quad dx$

Section C:

Integrals of the form $\int \frac{\mathbf{Ax+B}}{\mathbf{ax^2+bx+c}}\mathbf{dx}$ **and** $\int \frac{\mathbf{Ax+B}}{\sqrt{\mathbf{ax^2+bx+c}}}\mathbf{dx}$

25. $\int \frac{\mathbf{dx}}{\mathbf{1-9x^2}}$

26. $\int \frac{\mathbf{dx}}{\sqrt{\mathbf{16-9x^2}}}$

27. $\int \frac{\mathbf{dx}}{\sqrt{\mathbf{7+6x-x^2}}}$

28. $\int \frac{\mathbf{dx}}{\mathbf{x^2+2x+5}}$

29. $\int \frac{\mathbf{6+x}}{\sqrt{\mathbf{4-x^2}}}\mathbf{dx}$

30. $\int \frac{\mathbf{dx}}{\sqrt{\mathbf{x^2+4x}}}$

31. $\int \frac{\mathbf{dx}}{\sqrt{\mathbf{2x-x^2}}}$

32. $\int \frac{\mathbf{dx}}{\sqrt{\mathbf{x^2+4x}}}$

33. $\int \frac{\mathbf{dx}}{\mathbf{6x-x^2-7}}$

34. $\int \frac{\mathbf{(6x-7)\,dx}}{\mathbf{3x^2-7x+11}}$

35. $\int \frac{\mathbf{\sec x \tan x\, dx}}{\sqrt{\mathbf{\sec^2 x - 2\sec x}}}$

36. $\int \frac{\boldsymbol{sec^2 x}}{\boldsymbol{tan^2 x - 6 tan\, x + 10}}\boldsymbol{dx}$

37. $\int \frac{\boldsymbol{e^x}}{\boldsymbol{e^{2x}+2e^x+5}}$

Section D: Reduction Formulas.

Prove the following reduction formulas:

38. $\int \sin^n x \; dx \quad = I_n = \frac{-1}{n}\sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2}$

And hence find: $\int \boldsymbol{sin^6 x\, cosec^3 x\, dx}$

39. $\int \cos^n x \, dx = I_n = \frac{1}{n}\sin x \cos^{n-1} x + \frac{n-1}{n} I_{n-2}$

And hence find: $\int \boldsymbol{\cos^5 x \sec x \, dx}$

40. $\int \tan^n x \, dx = I_n = \frac{1}{n-1}\tan^{n-1} x - I_{n-2}$

And hence find: $\int \boldsymbol{\tan^5 x \, dx}$

41. $\int \sec^n x \, dx = I_n = \dfrac{1}{n-1}\tan x \sec^{n-2} x + \dfrac{n-2}{n-1} I_{n-2}$

And hence find: $\int \boldsymbol{\sec^3 x \, dx}$

Homework#3

Evaluate the following integrals:

1. $\int \sin^5 x \quad \cos^5 x \quad dx$
2. $\int \cos^9 x \quad \sin^3 x \quad dx$
3. $\int \sin^2 x \quad \cos^2 x \quad dx$
4. $\int \frac{\cos^3 x}{\sin^2 x} \quad dx$
5. $\int \cot^3 x \quad \sec^4 x \quad dx$
6. $\int \tan^3 x \quad \sec^6 x \quad dx$
7. $\int \tan^3 x \quad \sec^7 x \quad dx$
8. $\int \tan^6 x \quad \sec^4 x \quad dx$
9. $\int \tan^2 x \quad \cos^5 x \, dx$
10. $\int \tan x \quad \sin^2 x \, dx$
11. $\int \sin 7x \quad \cos 5x \quad dx$
12. $\int \sin 4x \quad \sin 2x \quad dx$
13. $\int \cos 5x \quad \cos 3x \quad dx$
14. $\int \cos 7x \quad \sin 2x \quad dx$

Evaluate the following integrals:

15. $\int \frac{x^3 + x}{\sqrt{1+x^2}} \quad dx$
16. $\int \frac{dx}{x\sqrt{9-x^2}}$
17. $\int x^2\sqrt{4-x^2} \quad dx$
18. $\int \sqrt{x^2+6} \quad dx$
19. $\int \frac{dx}{2x^2 - 4x + 4}$
20. $\int \frac{dx}{\sqrt{x^2+9}}$
21. $\int \frac{dz}{\sqrt{2z+z^2}}$
22. $\int \frac{dx}{\sqrt{2-3x-4x^2}}$
23. $\int \frac{dx}{6-x^2-8x}$
24. $\int \frac{dx}{\sqrt{x(3x+5)}}$

25. $\int \frac{(2x-6)dx}{x^2-8x}$

26. $\int \frac{(x+3)dx}{\sqrt{4x^2+4x+9}}$

27. $\int \frac{3x-1}{x^2-2x+2} \quad dx$

28. $\int \frac{\sec^2 x\, dx}{\tan^2 x - 4\tan x + 8}$

29. $\int \frac{10^x\, dx}{10^{2x} - 6(10)^x + 12}$

30. $\int \sqrt{x^2+6x} \quad dx$

31. $\int \sqrt{2x-x^2} \quad dx$

32. $\int \sin x \sqrt{1+\cos^2 x}\, dx$

33. $\int \frac{\cos 2x \;\; dx}{\sqrt{\sin^2 2x + 5 + 2\sin 2x}}$

34. $\int \frac{\sin 2x \;\; dx}{\sin^3 x + 9\sin x}$

35. $\int (1+\tan^2 x)^4\, dx$

36. $\int \frac{e^x\, dx}{\sqrt{e^{2x}+e^x+1}}$

37. $\int \frac{e^{2x}\, dx}{e^{4x}+2e^{2x}+5}$

Prove the following reduction formulas:

38. $\int \sin^n x\, dx = I_n = \frac{-1}{n}\sin^{n-1} x \cos x + \frac{n-1}{n} I_{n-2}$

And hence find: $\int \sin^6 x \quad dx$

39. $\int \cos^n x\, dx = I_n = \frac{1}{n}\sin x \cos^{n-1} x + \frac{n-1}{n} I_{n-2}$

And hence find: $\int \cos^6 x \sec^{-3} x \quad dx$

40. $\int \tan^n x\, dx = I_n = \frac{1}{n-1}\tan^{n-1} x - I_{n-2}$

And hence find: $\int \frac{dx}{\cot^4 x} \quad dx$

41. $\int \sec^n x\, dx = I_n = \frac{1}{n-1}\tan x \sec^{n-2} x + \frac{n-2}{n-1} I_{n-2}$

And hence find: $\int \frac{dx}{\cos^5 x}$

Sheet#4
Definite Integration and it's Application
Lecture Examples

Section A: "Definite integration"

1. $\int_{1}^{2} x^3 + 3x^2 + 1 \quad dx$

2. $\int_{0}^{\frac{\pi}{2}} \sin^2 x \quad dx$

3. $\int_{0}^{3} \sqrt{9 - x^2} \quad dx$

4. $\int_{0}^{3} \frac{2x - 1}{(x+1)(x+2)} \quad dx$

Section B: "Area & Volumes"

5. **Find the area of a figure bounded by the lines y = x, y=8 and x=0.**

6. **Find the area of a figure bounded by the parabola $y^2 = x+2$ and the line x = 2y-1.**

7. **A figure bounded by $y = x^3$, y=4 , y=1 and x=0.**
   a) **Find the area of this region.**
   b) **Compute the volume V_y of the solid generated by rotating this region one complete revolution about the y-axis.**

8. **A region in the xy-plan is bounded by the following curves:**
   $x^2 = 2-y$, x = 0 , y = x. (Consider the region in the 1st quadrant)
   a) **Find the area of this region.**
   b) **Compute the volume V_x of the solid generated by rotating this region one complete revolution about the x-axis.**

9. **A region in the xy-plan is bounded by the following curve**
   $y = \sin x$, $x = 0$, $x = \pi$.
   a) **Find the area of this region.**
   b) **Compute the volume V_x of the solid generated by rotating this region one complete revolution about the x-axis.**

Section C: "Length of a curve (arc length)"

The length of a curve $y = f(x)$ between x = a and x = b is defined by:

$$l = \int_a^b \sqrt{1 + (y')^2} \quad dx$$

While in parametric form $l = \int_a^b \sqrt{\left(\frac{dy}{d\theta}\right)^2 + \left(\frac{dx}{d\theta}\right)^2} \quad d\theta$

10. **Find the length of the curve** $y = x^{\frac{3}{2}}$ **for x in [0,1].**
11. **Find the length of the curve** $x = 3y^{\frac{3}{2}} - 1$ **for y in [0,4].**
12. **Find the length of the hypocycloid** $x = a\cos^3\theta$ **,** $y = a\sin^3\theta$ **for** $\theta \in \left[0, \pi/2\right]$**.**

Section D: " Average of a function"

13. **Find the average of the function** $y = x^2 + 2x + 1$ **from x =0 to x =2**
14. **Find the average of the function** $y = \cos^2 4x$ **from x = 0 to x =** $\frac{\pi}{4}$

Section E: "Numerical Integration"

Trapezoidal rule:

$$\int_a^b y(x)\, dx = \frac{h}{2}\left[(y_0 + y_n) + 2(y_1 + y_2 + \cdots\cdots + y_{n-1})\right]$$

Simpson's rule:

$$\int_a^b y(x)\, dx = \frac{h}{3}\left[(y_0 + y_n) + 2(y_2 + y_4 + \cdots + y_{n-2}) + 4(y_1 + y_3 + \cdots + y_{n-1})\right]$$

Note: step size h = (b-a)/n , n (no. of intervals) must be even for Simpson's rule, if n is not given it is assumed to be 6.

15. Compute $\int_0^2 \frac{dx}{4+x^2}$ by the Trapezoid rule and also Simpson's rule, then find the absolute error for both methods, and hence give an approx. value for π.

16. Compute $\int_1^4 \frac{dx}{x+1}$, by the Trapezoid rule and also Simpson's rule, then find the absolute error for both methods, and hence give an approx. value for Ln (2.5).

17. Compute $\int_0^1 e^{-x^2} dx$,using both the Trapezoid and Simpson's rules, take the number of intervals n=10. (answer Simpson = 0.746825, Trapezoid = 0.746211)

Sheet#4
Classroom exercises
Definite Integration and it's Application

Section A: "Definite integration"

1. $\int_0^{\frac{1}{2}} \frac{dx}{4+x^2}$

2. $\int_0^1 3^x e^x \, dx$

3. $\int_0^1 x \tan^{-1} x \, dx$

4. $\int_0^3 \frac{2x^2-3x-3}{(x+1)(x^2-2x+5)} \, dx$

Section B: "Area & Volumes"

5. **Find the area of the figure bounded by the x-axis, x=4 and y = x.**

6. **Find the area of the region bounded by $y=x^3$ and y=8 and the y-axis.**

7. **A region in the xy-plan is bounded by the following lines:**
   y = x, x +y=6 , y = 0
   i) **Find the area of this region.**
   ii) **Compute the volume V_y of the solid generated by rotating this region one complete revolution about the y-axis.**

8. **A region in the xy-plan is bounded by the following curve**
   $y= (x-2)^2$, y = x.
   a) **Find the area of this region.**
   b) **Compute the volume V_x of the solid generated by rotating this region one complete revolution about the x-axis.**

9. **A region in the xy-plan is bounded by the ellipse**

$$\frac{x^2}{16}+\frac{y^2}{9}=1$$

a) **Find the area of this region.**
b) **Compute the volume V_x of the solid generated by rotating this region one complete revolution about the x-axis.**

<u>Section C:</u> "Length of a curve (arc length)"

10. **Find the length of the curve** $y = \ln \sec x$ **for x in** $\left[\frac{\pi}{4},\frac{\pi}{3}\right]$.

11. **Find the length of the curve given by y = t^2+3 , x = 2t+2 from t =0 to t=4**

12. **Find the length of one arc of the cycloid** $x = a\theta - a\sin\theta$, $y = a - a\cos\theta$.

<u>Section D:</u> " Average of a function"

13. **Find the average of the function** $y = x^3 + x + 1$ **from x=0 to x=1**

14. **Find the average of the function** $y = \sin 3x$ **from x = 0 to x =** π

15. **Find the average of the function** $y = \sec^2 2x$ **from x = 0 to x =** $\frac{\pi}{2}$

<u>Section E:</u> "Numerical Integration"

16. **Compute** $\int_0^1 \frac{dx}{1+x^2}$ **by the Trapezoid rule and also Simpson's rule, then find the absolute error for both methods, and hence give an approx. value for** π**.**

17. **Compute** $\int_1^2 \frac{dx}{x}$ **by the Trapezoid rule and also Simpson's rule, then find the absolute error for both methods, and hence give an approx. value for Ln2.**

18. **Compute** $\int_0^1 \cos x^2 \quad dx$ **,using both Trapezoid and Simpson's rules.**

Homework#4

Definite integration:

1. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2 x \quad dx$

2. $\int_{3}^{4} \sqrt{3x^2 + 2} \quad x \quad dx$

3. $\int_{0}^{1/2} \frac{dx}{\sqrt{1 - 4x^2}}$

4. $\int_{0}^{\frac{\pi}{2}} x \quad \sin 2x \quad dx$

5. $\int_{0}^{\frac{\pi}{2}} \cos^3 x \quad \sin^2 x \quad dx$

6. $\int_{0}^{\frac{\pi}{4}} \tan^2 x \quad \sec^4 x \quad dx$

7. $\int_{0}^{\frac{\pi}{4}} \sec^3 x \quad dx$

Area & Volumes.

8. **A region in the xy-plan is bounded by the following curves:**
   $y = x$, $x + y = 6$, $x = 0$
   a) **Find the area of this region.**
   b) **Compute the volume V_x of the solid generated by rotating this region one complete revolution about the x-axis.**

9. **A region in the xy-plan is bounded by the following curves:**
   $y = (x-2)^2$, $y = x$.
   a) **Find the area of this region.**
   b) **Compute the volume V_x of the solid generated by rotating this region one complete revolution about the x-axis.**

10. **A region in the xy-plan is bounded by the circle**
    $x^2 + y^2 = 16$ (Consider only the region in the 1st quadrant)

i) **Find the area of this region.**
ii) **Compute the volume V_x of the solid generated by rotating this region one complete revolution about the x-axis.**

11. A region in the xy-plan is bounded by the circle

$$x^2 + y^2 = 9 \quad , \; y = x \text{ (Consider only the 1st quadrant)}$$

i) Find the area of this region.

ii) Compute the volume V_y of the solid generated by rotating this region one complete revolution about the y-axis.

12. A region in the xy-plan is bounded by the two curves

$$y^2=x, \; x^2=y$$

i) Find the area of this region.

ii) Compute the volume V_x of the solid generated by rotating this region one complete revolution about the x-axis.

13. (*this is an A-student question*)

A region in the xy-plan is bounded by : $y=2x$, $y=x$, $y = x^3$

(Consider only the region in the 1st quadrant)

i) Find the area of this region.

ii) Compute the volume V_x of the solid generated by rotating this region one complete revolution about the x-axis.

Length of a curve (arc length):

14. Find the length of the curve $y = 3x^2$ for x in [0,1].

15. Find the length of the curve $y = \frac{1}{3}\ln \sec 3x$ for x in $\left[0, \frac{\pi}{9}\right]$.

16. Find the length of one arc of the cycloid $x = \theta - \sin\theta$, $y = 1 - \cos\theta$.

17. Find the length of one arc of the hypocycloid $x = 4\sin\theta$, $y = 4\cos\theta$ for $\theta \in [0, \pi/2]$.

Average of a function:

18. Find the average of the function $z = y^3 - 3$ from y = 0 to y = 2

19. Find the average of the function $y = \sin^2 x$ from x = 0 to x = π

20. Find the average of the function $y = \cos^2 x + \sin x$ from x =0 to x = π

21. Find the average of the function $y = \tan^2 4x$ from x = 0 to x = $\frac{\pi}{2}$

Numerical Integration:

22. Compute $\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}}$ **by the Trapezoid rule and also Simpson's rule, then find the absolute error for both methods, and hence give an approx. value for** π**.**

23. Compute $\int_1^2 \frac{x\,dx}{x^2+1}$**, by the Trapezoid rule and also Simpson's rule, then find the absolute error for both methods, and hence give an approx. value for Ln (2.5)**

24. Compute $\int_1^6 \frac{4\,dx}{x+1}$**. by the Trapezoid rule and also Simpson's rule, then find the absolute error for both methods, and hence give an approx. value for Ln (3.5)**

25. Compute $\int_{\pi/2}^{\pi} \frac{\sin x}{x}\,dx$ **,using both Trapezoid and Simpson's rules.**

Sheet#5
Linear Algebra
Lecture Examples

Section A: "Matrix algebra"

Given:

$$A=\begin{pmatrix} 6 & -3 & 0 \\ -3 & 6 & -2 \\ 0 & -3 & 4 \end{pmatrix},\ B=\begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \end{pmatrix} \text{ And } C=\begin{pmatrix} 7 & 2 & 0 \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{pmatrix}$$

Find:

a) A+2B

b) A+AB+AC

c) A^t

d) AB

e) B^tA^t

f) A^{-1}

g) $(AB)^{-1}$

h) $(B^{-1}A^{-1})$

Section B: "systems of linear equations"

Solve each of the following using Gauss Elimination method, Cramer's method and the inverse matrix method:

1)
$$\begin{aligned} X + Y + Z &= 2 \\ X + 2Y - Z &= 0 \\ 2X - Z &= 1 \end{aligned}$$
(sol: x= 1 ,y = 0, z=1)

2)
$$\begin{aligned} 2X + Y + Z &= 8 \\ 4X + Y &= 11 \\ -2X + 2Y + Z &= 3 \end{aligned}$$
(sol: x= 2 ,y =3, z=1)

3)
$$\begin{aligned} 5Y + 7Z &= 2 \\ 6X + 9Y + 8Z &= 5 \\ 2X + 3Y + 3Z &= 2 \end{aligned}$$
(Sol: x= 1 ,y = -1, z =1)

4)
$$\begin{aligned} Y + Z + 2X &= 1 \\ 4X + Y &= -2 \\ Z - 2X + Y &= 7 \end{aligned}$$
(sol: x= -1 ,y = 2, z=1)

<u>Section C:</u> "Eigenvalues, Eigenvectors and Cayley-Hamilton theorem"

[1] Find the eigenvalues and eigenvectors of the given matrix:

a) $A=\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 4 & 0 & 1 \end{pmatrix}$

b) $A=\begin{pmatrix} 1 & 6 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

[2] Using Cayley-Hamilton theorem, find the inverse of the given matrix

a) $A=\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \\ 2 & 0 & -1 \end{pmatrix}$

b) $A=\begin{pmatrix} 0 & 5 & 7 \\ 6 & 9 & 8 \\ 2 & 3 & 3 \end{pmatrix}$

c) $A=\begin{pmatrix} 2 & 1 & 1 \\ 4 & 1 & 0 \\ -2 & 2 & 1 \end{pmatrix}$

d) $A=\begin{pmatrix} 2 & 1 & 1 \\ 4 & 1 & 0 \\ -2 & 1 & 1 \end{pmatrix}$

Sheet#5
Linear Algebra
Classroom Exercises

Section A: "Matrix algebra"

Given:

$$A=\begin{pmatrix}-3 & -2 & 0\\ -2 & 3 & -2\\ 0 & -6 & 5\end{pmatrix},\ B=\begin{pmatrix}1 & 1 & 0\\ 1 & -1 & 1\\ 0 & 3 & -2\end{pmatrix}\ \text{And}\ C=\begin{pmatrix}1 & 1 & 2\\ 1 & 2 & 3\\ 0 & 1 & 3\end{pmatrix}$$

Find:

a) AB

b) AC+AB+2B C

c) AI+2C-3B

d) B^tA^t

e) C^{-1}

f) $(C^t)^{-1}$

g) $(AB)^{-1}$

h) $(B^{-1}A^{-1})$

i) $(B^{-1}A^{-1})^{-1}$

Section B: "systems of linear equations"

Solve each of the following using Gauss Elimination method, Cramer's method and the inverse matrix method:

1)
$$\begin{aligned}2X + 2Y - Z &= -6\\ X \qquad - 2Z &= -3\\ -3X + Y + 4Z &= 2\end{aligned}$$
(sol: x= 1 ,y = -3, z=2)

2)
$$\begin{aligned}X + Y - Z &= 3\\ 2Z + 2Y &= 4\\ Y + 4X - 3Z &= -2\end{aligned}$$
(sol: x= -3 ,y =4, z=-2)

3)
$$\begin{aligned}X + 2Y + Z &= 3\\ 2X - 4Y \qquad &= 4\\ 5X - 2Y - 6Z &= 4\end{aligned}$$
(Sol: x= 2 ,y = 0, z=1)

4)
$$\begin{aligned}X - 2Y \qquad &= 3\\ 2X + Y + 2Z &= 5\\ 3X - Y - Z &= 2\end{aligned}$$
(sol: x= 1 ,y = -1, z=2)

<u>Section C:</u> "Eigenvalues, Eigenvectors and Cayley-Hamilton theorem"

Find the eigenvalues and eigenvectors of the given matrix:

1) $A=\begin{pmatrix} 5 & 0 & 0 \\ 0 & -5 & 0 \\ 5 & -1 & 0 \end{pmatrix}$

2) $A=\begin{pmatrix} -2 & 4 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

[2] Using Cayley-Hamilton theorem, find the inverse of the given matrix

a) $A=\begin{pmatrix} 2 & 2 & -1 \\ 1 & 0 & -2 \\ -3 & 1 & 4 \end{pmatrix}$

b) $A=\begin{pmatrix} 1 & 2 & 1 \\ 2 & -4 & 0 \\ 5 & -2 & -6 \end{pmatrix}$

c) $A=\begin{pmatrix} 1 & 1 & -1 \\ 0 & 2 & 2 \\ 4 & 1 & -3 \end{pmatrix}$

d) $A=\begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 2 \\ 3 & -1 & -1 \end{pmatrix}$

Homework#5

Given:

$$A=\begin{pmatrix}2 & 3 & 0\\ 5 & 2 & 1\\ 0 & 3 & 4\end{pmatrix}\ ,\ B=\begin{pmatrix}1 & 1 & 2\\ 1 & 1 & 3\\ 0 & 2 & 4\end{pmatrix}\ \textbf{and}\ C=\begin{pmatrix}-2 & -4 & -2\\ -6 & -2 & -4\\ 0 & -5 & -7\end{pmatrix}$$

Find:

a)	$BA+BC+B^2$	**d)**	$(B^t)^{-1}$	**g)**	$BA-B^{-1}$
b)	B^{-1}	**e)**	$(BA)^t$	**h)**	$(AB)^t(B^tA^t)^{-1}$
c)	A^t	**f)**	A^tB^t	**i)**	$B((A-B)B^{-1})$

Solve using Gauss Elimination method then resolve using Cramer's method:

1)
$$\begin{aligned} X - 2Y &= 7\\ 2X + 4Y + Z &= -3\\ -2X + Y + 2Z &= 9 \end{aligned}$$
(sol: x= 1 ,y = -3, z=7)

2)
$$\begin{aligned} X - Y &= 4\\ 2X + Y + Z &= 0\\ Y - 2X + 2Z &= -3 \end{aligned}$$
(sol: x= 1 ,y = -3, z=1)

3)
$$\begin{aligned} X - Y + Z &= 1\\ 2X + Y + Z &= 4\\ Y - 2X &= -1 \end{aligned}$$
(sol: x= 1 ,y = 1, z=1)

4)
$$\begin{aligned} -Y + 3Z &= 5\\ 2X + 5Z &= 10\\ 3X + 3Y - 4Z &= -5 \end{aligned}$$
(sol: x= 0 ,y = 1, z=2)

<u>Section C:</u> "Eigenvalues, Eigenvectors and Cayley-Hamilton theorem"

[1] Find the eigenvalues and eigenvectors of the given matrix:

a) $A=\begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 6 \\ 0 & 0 & -7 \end{pmatrix}$

b) $A=\begin{pmatrix} 1 & 0 & 0 \\ 1 & 3 & 0 \\ 1 & 1 & 3 \end{pmatrix}$

[2] Using Cayley-Hamilton theorem, find the inverse of the given matrix

a) $A=\begin{pmatrix} 1 & -2 & 0 \\ 2 & 4 & 1 \\ -2 & 1 & 2 \end{pmatrix}$

b) $A=\begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 1 \\ -2 & 1 & 0 \end{pmatrix}$

c) $A=\begin{pmatrix} 1 & -1 & 0 \\ 2 & 1 & 1 \\ -2 & 1 & 2 \end{pmatrix}$

d) $A=\begin{pmatrix} 0 & -1 & 3 \\ 2 & 0 & 5 \\ 3 & 3 & -4 \end{pmatrix}$

Appendix B
"Answers to H.W Problems"

H.W#1

1) $\frac{1}{2}\sqrt{x} - 8/\sqrt{x} - 2x + c$

2) $\frac{1}{5}e^{5x} + c$

3) $\frac{1}{7}\tan 7x + c$

4) tan(lnx) + c

5) $\frac{1}{5}$Ln sin(5x-7) +c

6) $\frac{1}{2}\tan^2 x + c$

7) $\ln^2 x + c$

8) $\frac{1}{3}(x^2+1)^{3/2} + c$

9) $\frac{1}{4}$ln sec4z – 4lnsin(z/4) +c

10) $\frac{2}{3}\sqrt{x^3+1} + c$

11) – cosecx +c

12) 2tanx + secx – x +c

13) $\sin^{-1}\ln x + c$

14) $2\sqrt{\tan x - 1} + c$

15) $\ln \sin^{-1} x + c$

16) ln ln lnx +c

17) $\frac{1}{2}$(tanx – x) +c

18) $2\sqrt{1+\sin^2 x} + c$

19) $\frac{a^{x^2}}{2\ln a} + c$

20) $\frac{1}{2}e^{x^2+4x+3} + c$

21) $\frac{-1}{9}(1-3x^2)^{-3/2} + c$

22) $\frac{1}{2}\ln\ln x^2 + c$

23) $\frac{1}{2}(\sin^{-1}x)^2 + c$

24) $\frac{1}{7}$ln(sec(7x-2) + tan(7x-2)) +c

25) $\frac{-1}{25}(5\tan x - 3)^{-5} + c$

26) ln tanx +c

27) $\frac{-1}{\pi}\cot \pi x + c$

28) $\frac{1}{6}(e^x+3)^6 + c$

29) $\frac{-1}{3}\operatorname{cosec} 3x + c$

30) $\sin^{-1}(x/3) + c$

31) $\tan^{-1} e^x + c$

32) $\frac{3}{50}\ln(25x^2+49) + \frac{2}{50}\tan^{-1}(5x/7) + c$

33) $\tan^{-1}\ln x + c$

34) x+c

H.W#2

1) $x \tan x - \ln \sec x + c$

2) $x \tan^{-1}x + \frac{1}{2}\ln(1+x^2) + c$

3) $x \ln(x-1) + x + \ln(x-1) + c$

4) $x\left(\frac{1}{2}x + \frac{1}{4}\sin 2x\right) - \left(\frac{1}{4}x^2 + \frac{1}{8}\cos 2x\right) + c$

5) $\frac{1}{5}x^2 e^{5x} - \frac{2}{25}x e^{5x} + \frac{2}{125}e^{5x} + c$

6) $x^2 \sin x + 2x\cos x - 2\sin x + c$

7) $\frac{1}{2}(x+4)e^{2x-3} - \frac{1}{4}e^{2x-3} + c$

8) $30 e^x \cos 3x + 10 e^x \sin 3x + c$

9) $\frac{1}{5}x^5 \ln x^4 - \frac{4}{25}x^5 + c$

10) $x \tan x - \ln \sec x - \frac{1}{2}x^2 + c$

11) $\frac{1}{\ln 3}x\, 3^x - \frac{1}{(\ln 3)^2}3^x + c$

12) $\frac{5}{2}x e^{2x} - \frac{5}{4}e^{2x} + c$

13) $\frac{1}{6}(1+x^2)^3 + c$

14) $2x + \frac{3}{2}x^2 + \frac{1}{2}\tan^{-1}(x/2) + \frac{1}{2}\ln(x^2+4) + c$

15) $\frac{1}{2}\ln(x-3) - \ln(x-2) + \frac{1}{2}\ln(x-1) + c$

16) $\frac{1}{4}\ln(x+1) - \frac{1}{3}\ln(x+2) + \frac{1}{12}\ln(x+5) + c$

17) $\frac{8}{5}\ln(x-7) + \frac{7}{5}\ln(x+3) + c$

18) $2\ln(x-1) + 2\tan^{-1}x - \ln(x^2+1) + c$

19) $\frac{1}{3}x^3 + \frac{1}{2}x^2 + 4x + 5\ln(x-2) - 3\ln(x+2) + 2\ln x + c$

20) $\ln(x-1) - \ln(x-3) + c$

21) $2\ln(x^2+4) + \tan^{-1}x - 3\ln(x+5) + c$

22) $\ln(x-1) + \ln(x+1) - \ln(x^2+1) + c$

23) $\frac{3}{2}x^2 + 2x + 6\ln(x-1) + \frac{6}{x} - 6\ln x + c$

24) $x + \frac{7}{2}\ln(x-2) + \frac{3}{2}\ln(x+2) + c$

25) $\ln(x-1) + (x-1)^{-1} - 2(x-1)^2 - \ln x + c$

26) $\tan^{-1}x - (x+1)^{-1} + c$

H.W#3

1) $\frac{1}{6}\sin^6 x - \frac{1}{4}\sin 8x + \frac{1}{10}\sin^{10}x + c$

2) $\frac{1}{12}\cos^{12}x - \frac{1}{10}\cos^{10}x + c$

3) $\frac{1}{16}x^2 + \frac{1}{128}\cos^{10}x + c$

4) $-\operatorname{cosec} x - x + c$

5) $\frac{-1}{2}(\tan x)^{-2} + \ln \tan x + c$

6) $\frac{1}{8}\sec^8 x - \frac{1}{6}\sec^6 x + c$

7) $\frac{1}{9}\sec^9 x - \frac{1}{7}\sec^7 x + c$

8) $\frac{1}{7}\tan^7 x - \frac{1}{9}\tan^9 x + c$

9) $\frac{1}{3}\sin^3 x - \frac{1}{5}\sin^5 x + c$

10) $\frac{1}{2}\tan^{-1}(x-1) + c$

11) $\ln\left|x+\sqrt{x^2+9}\right| + c$

12) $\ln\left|z+1+\sqrt{z^2+2z}\right| + c$

13) $\frac{1}{2}\sin^{-1}\left(\frac{2x+3/4}{7}\right) + c$

14) $\frac{1}{2\sqrt{22}}\ln\left|\frac{\sqrt{22}+(x+4)}{\sqrt{22}-(x+4)}\right| + c$

15) $\frac{1}{\sqrt{3}}\ln\left|(x+\frac{5}{6})+\sqrt{3x^2+5x}\right| + c$

16) $\ln(x^2-8x) + \frac{1}{4}\ln(x/(8-x)) + c$

17) $\frac{1}{4}\sqrt{4x^2+4x+9}$
$+2\ln\left|(x+1)+\sqrt{(x+1)^2+\frac{5}{4}}\right| + c$

18) $\frac{3}{2}\ln(x^2+2x+2) - 4\tan^{-1}(x+1) + c$

19) $\frac{1}{2}\tan^{-1}(\frac{1}{2}\tan x - 1) + c$

20) $\frac{1}{2}\tan^{-1}(\frac{1}{\sqrt{3}}(10^x - 3)) + c$

25) $\frac{-1}{2}(\cos x\sqrt{1+\cos^2 x} + \ln\left|\cos x+\sqrt{1+\cos^2 x}\right| + c$

26) $\frac{1}{2}(\ln x\sqrt{(\ln x-2)^2+1} + \ln\left|\ln x+\sqrt{(\ln x-2)^2+1}\right| + c$

27) $\frac{1}{2}(\ln\left|\sin 2x+\sqrt{(\sin 2x+1`)^2+4}\right| + c$

28) $\frac{2}{3}\tan^{-1}(\frac{\sin x}{3}) + c$

H.W#4

1) $\pi/2$

2) 21.932

3) $\pi/4$

4) $\pi/4$

5) 2/15

6) 8/15

7) $\frac{1}{2}[\sqrt{2}+\ln(1+\sqrt{2})]$

8) A = 9, $V_x = 72\pi$

9) A= 14/3 , $V_x = 72\pi/5$

10) A $= 4\pi$, $V_x = 128\pi/3$

11) A $= 9\pi/8$, $V_y = 9\sqrt{2}\pi$

12) A =1/3 , $V_x = 3\pi/10$

13) A =3/4, $V_x = 1.96452\pi$

14) $A = 6\pi$

15) $A = 6\pi$

16) $A = (5\pi/4 - 2)\, a^2$

17) $A = (2 - \pi/4)\, a^2$

18) $A = a^2\pi/4$

19) $A = 5\pi a^2/4$

20) $A = (\frac{\pi}{6} - \frac{\sqrt{3}}{2})\, a^2$

21) 3.09972

22) 0.43896

23) 8

24) 6

25) –1

26) ½

27) $\frac{2}{\pi} + \frac{1}{2}$

29) –1

H.W#5

a) $\begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 3 \\ 0 & 2 & 4 \end{pmatrix}$

b) $\frac{-1}{2}\begin{pmatrix} -2 & 0 & 1 \\ -4 & 4 & -1 \\ 2 & -2 & 0 \end{pmatrix}$

c) $\begin{pmatrix} 2 & 5 & 0 \\ 3 & 2 & 3 \\ 0 & 1 & 4 \end{pmatrix}$

d) $(B^{-1})^T$

e) $\begin{pmatrix} 7 & 7 & 10 \\ 11 & 14 & 16 \\ 9 & 13 & 18 \end{pmatrix}$

f) $(BA)^T$

g) $\begin{pmatrix} 6 & 11 & 19/2 \\ 5 & 16 & 25/2 \\ 11 & 15 & 18 \end{pmatrix}$

h) I

i) $\begin{pmatrix} 19 & -14 & 0 \\ 21 & -16 & 1/2 \\ 24 & -16 & -1 \end{pmatrix}$